

Grade 7
Unit 10
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

Use the given information to answer questions 1-2.

Lisa solved the equation $-2x - 3 = 5$. Her work is shown.

$$\begin{aligned} -2x - 3 + 3 &= 5 + 3 \\ -2x &= 8 \\ -2x \div -2 &= 8 \div -2 \\ x &= 4 \end{aligned}$$

1. Use substitution to verify if the answer is correct.

$$\begin{aligned} -2(4) - 3 & & -11 \neq 5 \text{ so } x \neq 4 \\ -8 - 3 &= -11 \end{aligned}$$

2. If correct, describe the properties of equality applied. If incorrect, describe the error(s) made.

$$8 \div (-2) = -4$$

Use the given information to answer questions 3-4.

Zoey solved the equation $3m + 5 = 8$. Her work is shown.

$$\begin{aligned} 3m + 5 - 5 &= 8 - 5 \\ 3m &= -3 \\ 3m \div 3 &= -3 \div 3 \\ m &= -1 \end{aligned}$$

3. Use substitution to verify if the answer is correct.

$$\begin{aligned} 3(-1) + 5 & & 2 \neq 8 \text{ so } m \neq -1 \\ -3 + 5 &= 2 \end{aligned}$$

4. If correct, describe the properties of equality applied. If incorrect, describe the error(s) made.

$$8 - 5 = 3 \text{ not } -3$$

5. A taxi charges \$1.75 plus a fee of \$0.75 per mile traveled. The cost of your trip was \$14.50. The equation $\$1.75 + \$0.75x = \$14.50$ can be used to represent the situation. Solve. Show your work.

$$\begin{aligned} 1.75 + 0.75x &= 14.50 \\ -1.75 & \quad -1.75 \\ \hline 0.75x &= 12.75 \\ \frac{0.75x}{0.75} &= \frac{12.75}{0.75} \\ x &= 17 \end{aligned}$$

6. Explain what the solution means in the context of question 5.
For \$14.50, you can travel 17 miles.

7. Rachael and George go ziplining. It costs a flat rate of \$75 and an additional \$5.50 for each line you cross. It cost \$124.50 per person to complete the course. The equation $\$75 + \$5.50z = \$124.50$ can be used to represent the situation. Solve. Show your work.

$$\begin{array}{rcl}
 75 + 5.5z & = & 124.50 \\
 -75 & & -75 \\
 \hline
 5.5z & = & 49.5 \\
 \frac{5.5}{5.5} & & \frac{49.5}{5.5} \\
 \hline
 z & = & 9
 \end{array}$$

8. Explain what the solution means in the context of question 7.

There were 9 lines crossed while ziplining.

9. Solve $9.2 = 6.1 + \frac{h}{6}$ using the properties of equality. Use substitution to verify your solution.

$$\begin{array}{rcl}
 9.2 & = & 6.1 + \frac{h}{6} \\
 -6.1 & & -6.1 \\
 \hline
 3.1 & = & \frac{h}{6} \\
 6 \cdot 3.1 & = & \frac{h}{6} \cdot 6 \\
 18.6 & = & h
 \end{array}$$

$$\begin{array}{l}
 \text{Check: } \frac{18.6}{6} + 6.1 \\
 3.1 + 6.1 = 9.2
 \end{array}$$

10. Solve $\frac{t-50}{4} = 400$ using the properties of equality. Use substitution to verify your solution is correct.

$$\begin{array}{rcl}
 4 \cdot \frac{t-50}{4} & = & 400 \cdot 4 \\
 t-50 & = & 1600 \\
 +50 & & +50 \\
 \hline
 t & = & 1650
 \end{array}$$

$$\begin{array}{rcl}
 \text{Check:} & & \\
 1650 - 50 & & \\
 \hline
 1600 & & \\
 \div 4 & & \\
 \hline
 400 & &
 \end{array}$$